

SVM and Kernel machine

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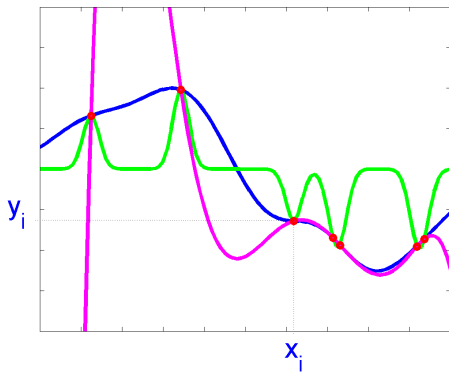
Plan

1 Kernel machines

- Non sparse kernel machines
- sparse kernel machines: SVM
- Sparse kernel machines for regression: SVR
- practical SVM

Interpolation splines

find out $f \in \mathcal{H}$ such that $f(\mathbf{x}_i) = y_i, \quad i = 1, \dots, n$



It is an ill posed problem

Interpolation splines: minimum norm interpolation

$$\begin{cases} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{such that} & f(\mathbf{x}_i) = y_i, \quad i = 1, \dots, n \end{cases}$$

The lagrangian (α_i Lagrange multipliers)

$$L(f, \alpha) = \frac{1}{2} \|f\|^2 - \sum_{i=1}^n \alpha_i (f(\mathbf{x}_i) - y_i)$$

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optimality for f : $\nabla_f L(f, \alpha) = 0 \quad \Leftrightarrow \quad f(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}_i, \mathbf{x})$

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dual formulation (remove f from the lagrangian):

$$Q(\alpha) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i y_i \quad \text{solution: } \max_{\alpha \in \mathbb{R}^n} Q(\alpha)$$

$$K\alpha = y$$

Representer theorem

Theorem (Representer theorem)

Let $\mathcal{H}$ be a RKHS with kernel $k(s, t)$. Let ℓ be a function from $\mathcal{X}$ to $\mathbb{R}$ (loss function) and Φ a non decreasing function from $\mathbb{R}$ to $\mathbb{R}$. If there exists a function f^* minimizing:

$$f^* = \underset{f \in \mathcal{H}}{\operatorname{argmin}} \sum_{i=1}^n \ell(y_i, f(\mathbf{x}_i)) + \Phi(\|f\|_{\mathcal{H}}^2)$$

then there exists a vector $\alpha \in \mathbb{R}^n$ such that:

$$f^*(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}, \mathbf{x}_i)$$

it can be generalized to the semi parametric case: $+ \sum_{j=1}^m \beta_j \phi_j(\mathbf{x})$

Elements of a proof

- ① $\mathcal{H}_s = \text{span}\{k(., \mathbf{x}_1), \dots, k(., \mathbf{x}_i), \dots, k(., \mathbf{x}_n)\}$
- ② orthogonal decomposition: $\mathcal{H} = \mathcal{H}_s \oplus \mathcal{H}_\perp \Rightarrow \forall f \in \mathcal{H}; f = f_s + f_\perp$
- ③ pointwise evaluation decomposition

$$\begin{aligned} f(\mathbf{x}_i) &= f_s(\mathbf{x}_i) + f_\perp(\mathbf{x}_i) \\ &= \langle f_s(.), k(., \mathbf{x}_i) \rangle_{\mathcal{H}} + \underbrace{\langle f_\perp(.), k(., \mathbf{x}_i) \rangle_{\mathcal{H}}}_{=0} \\ &= f_s(\mathbf{x}_i) \end{aligned}$$

- ④ norm decomposition
- ⑤ decompose the global cost

$$\|f\|_{\mathcal{H}}^2 = \|f_s\|_{\mathcal{H}}^2 + \underbrace{\|f_\perp\|_{\mathcal{H}}^2}_{\geq 0} \geq \|f_s\|_{\mathcal{H}}^2$$

$$\begin{aligned} \sum_{i=1}^n \ell(y_i, f(\mathbf{x}_i)) + \Phi(\|f\|_{\mathcal{H}}^2) &= \sum_{i=1}^n \ell(y_i, f_s(\mathbf{x}_i)) + \Phi(\|f_s\|_{\mathcal{H}}^2 + \|f_\perp\|_{\mathcal{H}}^2) \\ &\geq \sum_{i=1}^n \ell(y_i, f_s(\mathbf{x}_i)) + \Phi(\|f_s\|_{\mathcal{H}}^2) \end{aligned}$$

⑥

$\underset{f \in \mathcal{H}}{\operatorname{argmin}} = \underset{f \in \mathcal{H}_s}{\operatorname{argmin}}$

Smoothing splines

introducing the error (the slack) $\xi = f(x_i) - y_i$

$$(S) \quad \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + \frac{1}{2\lambda} \sum_{i=1}^n \xi_i^2 \\ \text{such that} & f(x_i) = y_i + \xi_i, \quad i = 1, n \end{array} \right.$$

three equivalent definitions

$$\begin{aligned} (S') \quad & \min_{f \in \mathcal{H}} \quad \frac{1}{2} \sum_{i=1}^n (f(x_i) - y_i)^2 + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 \\ & \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{such that} & \sum_{i=1}^n (f(x_i) - y_i)^2 \leq C' \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} & \sum_{i=1}^n (f(x_i) - y_i)^2 \\ \text{such that} & \|f\|_{\mathcal{H}}^2 \leq C'' \end{array} \right. \end{aligned}$$

using the representer theorem

$$(S'') \quad \min_{\alpha \in \mathbb{R}^n} \quad \frac{1}{2} \|K\alpha - \mathbf{y}\|^2 + \frac{\lambda}{2} \alpha^\top K \alpha$$

solution:

$$(S) \Leftrightarrow (S') \Leftrightarrow (S'') \Leftrightarrow (K + \lambda I)\alpha = \mathbf{y}$$

$\neq$ ridge regression:

$$\min_{\alpha \in \mathbb{R}^n} \quad \frac{1}{2} \|K\alpha - \mathbf{y}\|^2 + \frac{\lambda}{2} \alpha^\top \alpha$$

Kernel logistic regression

inspiration: the Bayes rule

$$D(\mathbf{x}) = \text{sign}(f(\mathbf{x}) + \alpha_0) \implies \log \left(\frac{\mathbb{P}(Y=1|\mathbf{x})}{\mathbb{P}(Y=-1|\mathbf{x})} \right) = f(\mathbf{x}) + \alpha_0$$

probabilities:

$$\mathbb{P}(Y = 1|\mathbf{x}) = \frac{\exp^{f(\mathbf{x})+\alpha_0}}{1 + \exp^{f(\mathbf{x})+\alpha_0}} \quad \mathbb{P}(Y = -1|\mathbf{x}) = \frac{1}{1 + \exp^{f(\mathbf{x})+\alpha_0}}$$

Rademacher distribution

$$\mathcal{L}(x_i, y_i, f, \alpha_0) = \mathbb{P}(Y = 1|\mathbf{x}_i)^{\frac{y_i+1}{2}} (1 - \mathbb{P}(Y = 1|\mathbf{x}_i))^{\frac{1-y_i}{2}}$$

penalized likelihood

$$\begin{aligned} J(f, \alpha_0) &= - \sum_{i=1}^n \log(\mathcal{L}(x_i, y_i, f, \alpha_0)) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 \\ &= \sum_{i=1}^n \log \left(1 + \exp^{-y_i(f(\mathbf{x}_i) + \alpha_0)} \right) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 \end{aligned}$$

Kernel logistic regression (2)

$$(\mathcal{R}) \quad \begin{cases} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + \frac{1}{\lambda} \sum_{i=1}^n \log(1 + \exp^{-\xi_i}) \\ \text{with} & \xi_i = y_i (f(\mathbf{x}_i) + \alpha_0), \quad i = 1, n \end{cases}$$

Representer theorem

$$J(\alpha, \alpha_0) = \mathbb{1}^\top \log \left(\mathbb{I} + \exp^{\text{diag}(\mathbf{y}) K \alpha + \alpha_0 \mathbf{y}} \right) + \frac{\lambda}{2} \alpha^\top K \alpha$$

gradient vector and Hessian matrix:

$$\nabla_{\alpha} J(\alpha, \alpha_0) = K(\mathbf{y} - (2\mathbf{p} - \mathbb{1})) + \lambda K \alpha$$

$$H_{\alpha} J(\alpha, \alpha_0) = K \text{diag}(\mathbf{p}(\mathbb{I} - \mathbf{p})) K + \lambda K$$

solve the problem using Newton iterations

$$\alpha^{\text{new}} = \alpha^{\text{old}} + (K \text{diag}(\mathbf{p}(\mathbb{I} - \mathbf{p})) K + \lambda K)^{-1} K(\mathbf{y} - (2\mathbf{p} - \mathbb{1}) + \lambda \alpha)$$

Let's summarize

- pros
 - ▶ Universality
 - ▶ from $\mathcal{H}$ to $\mathbb{R}^n$ using the representer theorem
 - ▶ no (explicit) curse of dimensionality
- splines $\mathcal{O}(n^3)$ (can be reduced to $\mathcal{O}(n^2)$)
- logistic regression $\mathcal{O}(kn^3)$ (can be reduced to $\mathcal{O}(kn^2)$)
- no scalability!

sparsity comes to the rescue!

Roadmap

1 Kernel machines

- Non sparse kernel machines
- **sparse kernel machines: SVM**
- Sparse kernel machines for regression: SVR
- practical SVM

SVM: the separable case (no noise)

$$\left\{ \begin{array}{ll} \max_{f, \alpha_0} & m \\ \text{with} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq m \\ \text{and} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 = 1 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{ll} \min_{f, \alpha_0} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{with} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq 1 \end{array} \right.$$

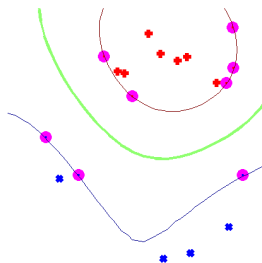
3 ways to represent function f

$$\underbrace{f(\mathbf{x})}_{\text{in the RKHS } \mathcal{H}} = \underbrace{\sum_{j=1}^d w_j \phi_j(\mathbf{x})}_{d \text{ features}} = \underbrace{\sum_{i=1}^n \alpha_i y_i k(\mathbf{x}, \mathbf{x}_i)}_{n \text{ data points}}$$

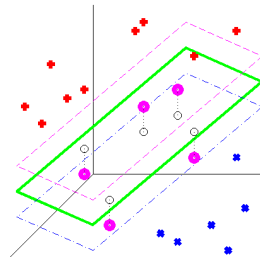
$$\left\{ \begin{array}{ll} \min_{\mathbf{w}, \alpha_0} & \frac{1}{2} \|\mathbf{w}\|_{\mathbb{R}^d}^2 = \frac{1}{2} \mathbf{w}^\top \mathbf{w} \\ \text{with} & y_i (\mathbf{w}^\top \phi(\mathbf{x}_i) + \alpha_0) \geq 1 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{ll} \min_{\alpha, \alpha_0} & \frac{1}{2} \alpha^\top K \alpha \\ \text{with} & y_i (\alpha^\top K(:, i) + \alpha_0) \geq 1 \end{array} \right.$$

using relevant features...

a data point becomes a function $\mathbf{x} \rightarrow k(\mathbf{x}, \bullet)$



input space representation: $\mathbf{x}$



feature space: $k(\mathbf{x}, \cdot)$

Representer theorem for SVM

$$\begin{cases} \min_{f, \alpha_0} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{with} & y_i(f(\mathbf{x}_i) + \alpha_0) \geq 1 \end{cases}$$

Lagrangian

$$L(f, \alpha_0, \alpha) = \frac{1}{2} \|f\|_{\mathcal{H}}^2 - \sum_{i=1}^n \alpha_i (y_i(f(\mathbf{x}_i) + \alpha_0) - 1) \quad \alpha \geq 0$$

optimality condition: $\nabla_f L(f, \alpha_0, \alpha) = 0 \Leftrightarrow f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i k(\mathbf{x}_i, \mathbf{x})$

Eliminate f from L :

$$\begin{cases} \|f\|_{\mathcal{H}}^2 = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) \\ \sum_{i=1}^n \alpha_i y_i f(\mathbf{x}_i) = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) \end{cases}$$

$$Q(\alpha_0, \alpha) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) - \sum_{i=1}^n \alpha_i (y_i \alpha_0 - 1)$$

Dual formulation for SVM

the intermediate function

$$Q(\alpha_0, \alpha) = -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) - \alpha_0 \left(\sum_{i=1}^n \alpha_i y_i \right) + \sum_{i=1}^n \alpha_i$$

$$\max_{\alpha} \min_{\alpha_0} Q(\alpha_0, \alpha)$$

α_0 can be seen as the Lagrange multiplier of the following (balanced) constraint $\sum_{i=1}^n \alpha_i y_i = 0$ which is also the optimality KKT condition on α_0

Dual formulation

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbf{R}^n} & -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i \\ \text{such that} & \sum_{i=1}^n \alpha_i y_i = 0 \\ \text{and} & 0 \leq \alpha_i, \quad i = 1, n \end{array} \right.$$

SVM dual formulation

Dual formulation

$$\left\{ \begin{array}{l} \max_{\alpha \in \mathbb{R}^n} \quad -\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j k(\mathbf{x}_i, \mathbf{x}_j) + \sum_{i=1}^n \alpha_i \\ \text{with} \quad \sum_{i=1}^n \alpha_i y_i = 0 \quad \text{and} \quad 0 \leq \alpha_i, \quad i = 1, n \end{array} \right.$$

The dual formulation gives a quadratic program (QP)

$$\left\{ \begin{array}{l} \min_{\alpha \in \mathbb{R}^n} \quad \frac{1}{2} \alpha^\top G \alpha - \mathbf{1}^\top \alpha \\ \text{with} \quad \alpha^\top \mathbf{y} = 0 \quad \text{and} \quad 0 \leq \alpha \end{array} \right.$$

with $G_{ij} = y_i y_j k(\mathbf{x}_i, \mathbf{x}_j)$

with the linear kernel $f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i (\mathbf{x}^\top \mathbf{x}_i) = \sum_{j=1}^d \beta_j x_j$
when d is small wrt. n primal may be interesting.

the general case: C-SVM

Primal formulation

$$(\mathcal{P}) \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \|f\|^2 + \frac{C}{p} \sum_{i=1}^n \xi_i^p \\ \text{such that} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, n \end{array} \right.$$

C is the *regularization path* parameter (to be tuned)

$p = 1$, L_1 SVM

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top G \alpha + \alpha^\top \mathbb{I} \\ \text{such that} & \alpha^\top \mathbf{y} = 0 \text{ and } 0 \leq \alpha_i \leq C \quad i = 1, n \end{array} \right.$$

$p = 2$, L_2 SVM

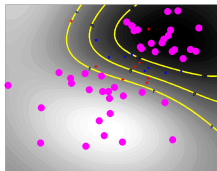
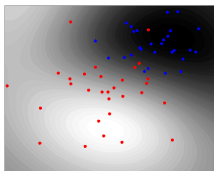
$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top (G + \frac{1}{C} I) \alpha + \alpha^\top \mathbb{I} \\ \text{such that} & \alpha^\top \mathbf{y} = 0 \text{ and } 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

the regularization path: is the set of solutions $\alpha(C)$ when C varies

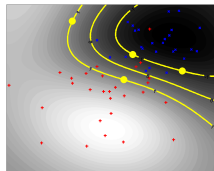
Data groups: illustration

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}, \mathbf{x}_i) + \alpha_0$$

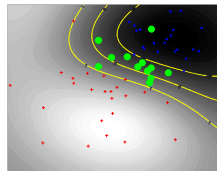
$$D(\mathbf{x}) = \text{sign}(f(\mathbf{x}))$$



useless data
well classified
 $\alpha = 0$



important data
support
 $0 < \alpha < C$



suspicious data
 $\alpha = C$

the regularization path: is the set of solutions $\alpha(C)$ when C varies

The importance of being support

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i k(\mathbf{x}_i, \mathbf{x})$$

data point	α	constraint value	set
$\mathbf{x}_i$ <i>useless</i>	$\alpha_i = 0$	$y_i (f(\mathbf{x}_i) + \alpha_0) > 1$	I_0
$\mathbf{x}_i$ <i>support</i>	$0 < \alpha_i < C$	$y_i (f(\mathbf{x}_i) + \alpha_0) = 1$	I_α
$\mathbf{x}_i$ <i>suspicious</i>	$\alpha_i = C$	$y_i (f(\mathbf{x}_i) + \alpha_0) < 1$	I_C

Table: When a data point is « support » it lies exactly on the margin.

here lies the efficiency of the algorithm (and its complexity)!

sparsity: $\alpha_i = 0$

The active set method for SVM (1)

$$\left\{ \begin{array}{ll} \min_{\alpha \in \mathbb{R}^n} & \frac{1}{2} \alpha^\top G \alpha - \alpha^\top \mathbb{I} \\ \text{such that} & \alpha^\top \mathbf{y} = 0 \quad i = 1, n \\ \text{and} & 0 \leq \alpha_i \quad i = 1, n \end{array} \right. \quad \left\{ \begin{array}{l} G\alpha - \mathbb{I} - \beta + b\mathbf{y} = 0 \\ \alpha^\top \mathbf{y} = 0 \\ 0 \leq \alpha_i \quad i = 1, n \\ 0 \leq \beta_i \quad i = 1, n \\ \alpha_i \beta_i = 0 \quad i = 1, n \end{array} \right.$$

G_a	$G_i^\top$
G_i	G_0

$$\begin{array}{c} \alpha_a \\ 0 \end{array} - \begin{array}{c} 1 \\ 1 \end{array} - \begin{array}{c} 0 \\ \beta_0 \end{array} + b \begin{array}{c} \mathbf{y}_a \\ \mathbf{y}_0 \end{array} = \begin{array}{c} 0 \\ 0 \end{array}$$

$G \qquad \alpha - \mathbb{I} - \beta + b \mathbf{y} = 0$

- (1) $G_a \alpha_a - \mathbb{I}_a + b \mathbf{y}_a = 0$
- (2) $G_i \alpha_a - \mathbb{I}_0 - \beta_0 + b \mathbf{y}_0 = 0$

- ① solve (1) (find α together with b)
- ② if $\alpha < 0$ move it from l_α to l_0
goto 1
- ③ else solve (2)
if $\beta < 0$ move it from l_0 to l_α
goto 1

The active set method for SVM (2)

Function $(\alpha, b, l_\alpha) \leftarrow \text{Solve_QP_Active_Set}(G, \mathbf{y})$

% Solve $\min_{\alpha} \quad 1/2 \alpha^\top G \alpha - \mathbf{1}^\top \alpha$
% s.t. $0 \leq \alpha$ and $\mathbf{y}^\top \alpha = 0$

$(l_\alpha, l_0, \alpha) \leftarrow \text{initialization}$

while The_optimal_is_not_reached **do**

$(\alpha, b) \leftarrow \text{solve} \begin{cases} G_a \alpha_a - \mathbf{1}_a + b \mathbf{y}_a = 0 \\ \mathbf{y}_a^\top \alpha_a \end{cases}$

if $\exists i \in l_\alpha$ such that $\alpha_i < 0$ **then**

$\alpha \leftarrow \text{projection}(\alpha_a, \alpha)$

move i from l_α to l_0

else if $\exists j \in l_0$ such that $\beta_j < 0$ **then**

use $\beta_0 = \mathbf{y}_0(K_i \alpha_a + b \mathbf{1}_0) - \mathbf{1}_0$

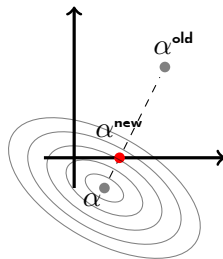
move j from l_0 to l_α

else

The_optimal_is_not_reached $\leftarrow \text{FALSE}$

end if

end while



Projection step of the active constraints algorithm

```
d = alpha - alphaold;  
alpha = alpha + t * d;
```

Caching Strategy

Save space and computing time by computing only the needed parts of kernel matrix G

Two more ways to derivate SVM

Using the hinge loss

$$\min_{f \in \mathcal{H}, \alpha_0 \in \mathbb{R}} \frac{1}{p} \sum_{i=1}^n \max(0, 1 - y_i(f(\mathbf{x}_i) + \alpha_0))^p + \frac{1}{2C} \|f\|^2$$

Minimizing the distance between the convex hulls

$$\left\{ \begin{array}{l} \min_{\alpha} \quad \|u - v\|_{\mathcal{H}}^2 \\ \text{with} \quad u(\mathbf{x}) = \sum_{\{i|y_i=1\}} \alpha_i k(\mathbf{x}_i, \mathbf{x}), \quad v(\mathbf{x}) = \sum_{\{i|y_i=-1\}} \alpha_i k(\mathbf{x}_i, \mathbf{x}) \\ \text{and} \quad \sum_{\{i|y_i=1\}} \alpha_i = 1, \quad \sum_{\{i|y_i=-1\}} \alpha_i = 1, \quad 0 \leq \alpha_i \quad i = 1, n \end{array} \right.$$

$$f(\mathbf{x}) = \frac{2}{\|u - v\|_{\mathcal{H}}^2} (u(\mathbf{x}) - v(\mathbf{x})) \quad \text{and} \quad \alpha_0 = \frac{\|u\|_{\mathcal{H}}^2 - \|v\|_{\mathcal{H}}^2}{\|u - v\|_{\mathcal{H}}^2}$$

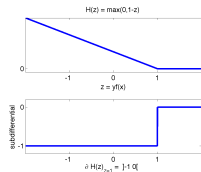
the regularization path: is the set of solutions $\alpha(C)$ when C varies

Regularization path for SVM

$$\min_{f \in \mathcal{H}} \sum_{i=1}^n \max(1 - y_i f(\mathbf{x}_i), 0) + \frac{\lambda_o}{2} \|f\|_{\mathcal{H}}^2$$

I_α is the set of support vectors s.t. $y_i f(\mathbf{x}_i) = 1$;

$$\partial_f J(f) = \sum_{i \in I_\alpha} \alpha_i y_i K(\mathbf{x}_i, \bullet) - \sum_{i \in I_1} y_i K(\mathbf{x}_i, \bullet) + \lambda_o f(\bullet) \quad \text{with} \quad \alpha_i \in \partial H(1) =]-1, 0[$$



Regularization path for SVM

$$\min_{f \in \mathcal{H}} \sum_{i=1}^n \max(1 - y_i f(\mathbf{x}_i), 0) + \frac{\lambda_o}{2} \|f\|_{\mathcal{H}}^2$$

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Let λ_n a value close enough to λ_o to keep the sets I_0, I_α and I_C unchanged

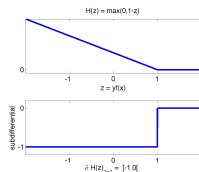
In particular at point $\mathbf{x}_j \in I_\alpha$ ($f_o(\mathbf{x}_j) = f_n(\mathbf{x}_j) = y_j$) : $\partial_f J(f)(\mathbf{x}_j) = 0$

$$\frac{\sum_{i \in I_\alpha} \alpha_{io} y_i K(\mathbf{x}_i, \mathbf{x}_j)}{\sum_{i \in I_\alpha} \alpha_{in} y_i K(\mathbf{x}_i, \mathbf{x}_j)} = \frac{\sum_{i \in I_1} y_i K(\mathbf{x}_i, \mathbf{x}_j) - \lambda_o y_j}{\sum_{i \in I_1} y_i K(\mathbf{x}_i, \mathbf{x}_j) - \lambda_n y_j}$$

$$G(\alpha_n - \alpha_o) = (\lambda_o - \lambda_n) \mathbf{y} \quad \text{avec} \quad G_{ij} = y_i K(\mathbf{x}_i, \mathbf{x}_j)$$

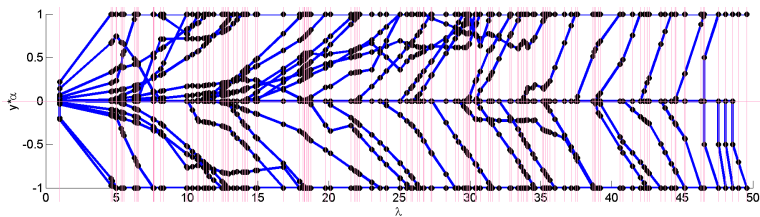
$$\alpha_n = \alpha_o + (\lambda_o - \lambda_n) \mathbf{w}$$

$$\mathbf{w} = (G)^{-1} \mathbf{y}$$

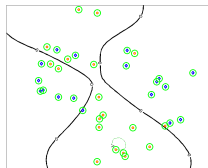
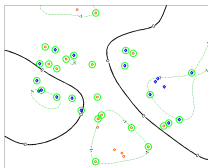
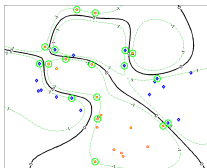


Example of regularization path

$$\alpha_i \in]-1, 0[\quad y_i \alpha_i \in]-1, -1[\quad \lambda = \frac{1}{C}$$



α_i estimation **and** data selection



How to choose ℓ and P to get linear *regularization path*?

the *path* is piecewise linear $\Leftrightarrow$ one is piecewise quadratic
and the other is piecewise linear

the convex case [Rosset & Zhu, 07]

$$\min_{\beta \in \mathbb{R}^d} \ell(\beta) + \lambda P(\beta)$$

① piecewise linearity: $\lim_{\varepsilon \rightarrow 0} \frac{\beta(\lambda + \varepsilon) - \beta(\lambda)}{\varepsilon} = \text{constant}$

② optimality

$$\begin{aligned}\nabla \ell(\beta(\lambda)) + \lambda \nabla P(\beta(\lambda)) &= 0 \\ \nabla \ell(\beta(\lambda + \varepsilon)) + (\lambda + \varepsilon) \nabla P(\beta(\lambda + \varepsilon)) &= 0\end{aligned}$$

③ Taylor expansion

$$\lim_{\varepsilon \rightarrow 0} \frac{\beta(\lambda + \varepsilon) - \beta(\lambda)}{\varepsilon} = [\nabla^2 \ell(\beta(\lambda)) + \lambda \nabla^2 P(\beta(\lambda))]^{-1} \nabla P(\beta(\lambda))$$

$$\nabla^2 \ell(\beta(\lambda)) = \text{constant} \quad \text{and} \quad \nabla^2 P(\beta(\lambda)) = 0$$

Problems with Piecewise linear regularization path

L	P	<i>regression</i>	<i>classification</i>	<i>clustering</i>
L_2	L_1	Lasso/LARS	L1 L2 SVM	PCA L1
L_1	L_2	SVR	SVM	OC SVM
L_1	L_1	L1 LAD Danzig Selector	L1 SVM	

Table: example of piecewise linear regularization path algorithms.

$$P: L_p = \sum_{j=1}^d |\beta_j|^p$$

$$L: L_p: |f(\mathbf{x}) - y|^p \quad \text{hinge } (yf(\mathbf{x}) - 1)_+^p$$

$$\varepsilon\text{-insensitive} \quad \begin{cases} 0 & \text{if } |f(\mathbf{x}) - y| < \varepsilon \\ |f(\mathbf{x}) - y| - \varepsilon & \text{else} \end{cases}$$

$$\text{Huber's loss:} \quad \begin{cases} |f(\mathbf{x}) - y|^2 & \text{if } |f(\mathbf{x}) - y| < t \\ 2t|f(\mathbf{x}) - y| - t^2 & \text{else} \end{cases}$$

K-Lasso (Kernel Basis pursuit)

The Kernel Lasso

$$(\mathcal{S}_1) \quad \left\{ \min_{\alpha \in \mathbb{R}^n} \quad \frac{1}{2} \|K\alpha - \mathbf{y}\|^2 + \lambda \sum_{i=1}^n |\alpha_i| \right.$$

- Typical parametric quadratic program (pQP) with $\alpha_i = 0$
- Piecewise linear regularization path

The dual:

$$(\mathcal{D}_1) \quad \left\{ \begin{array}{ll} \min_{\alpha} & \frac{1}{2} \|K\alpha\|^2 \\ \text{such that} & K^\top(K\alpha - \mathbf{y}) \leq t \end{array} \right.$$

- The K-Danzig selector can be treated the same way
- require to compute $K^\top K$ - no more function f !

Support vector regression (SVR)

Lasso's dual adaptation:

$$\left\{ \begin{array}{ll} \min_{\alpha} & \frac{1}{2} \|K\alpha\|^2 \\ \text{s. t.} & K^\top (K\alpha - \mathbf{y}) \leq t \end{array} \right. \quad \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 \\ \text{s. t.} & |f(\mathbf{x}_i) - y_i| \leq t, \quad i = 1, n \end{array} \right.$$

The support vector regression introduce slack variables

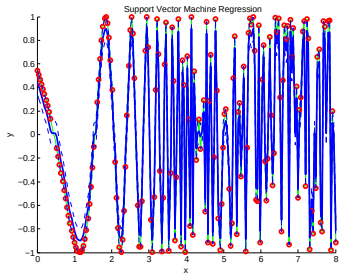
$$(SVR) \quad \left\{ \begin{array}{ll} \min_{f \in \mathcal{H}} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + C \sum |\xi_i| \\ \text{such that} & |f(\mathbf{x}_i) - y_i| \leq t + \xi_i \quad 0 \leq \xi_i \quad i = 1, n \end{array} \right.$$

- a typical **multi** parametric quadratic program (mpQP)
- piecewise linear regularization path

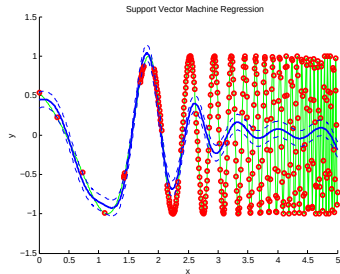
$$\alpha(C, t) = \alpha(C_0, t_0) + \left(\frac{1}{C} - \frac{1}{C_0}\right)\mathbf{u} + \frac{1}{C_0}(t - t_0)\mathbf{v}$$

- 2d Pareto's front (the tube width and the regularity)

Support vector regression illustration



C large



C small

- there exists other formulations such as LP SVR...

ν -SVM and other formulations...

$$\nu \in [0, 1]$$

$$(\nu) \left\{ \begin{array}{ll} \min_{f, \alpha_0, \xi, m} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + \frac{1}{np} \sum_{i=1}^n \xi_i^p - \nu m \\ \text{with} & y_i (f(\mathbf{x}_i) + \alpha_0) \geq m - \xi_i, \quad i = 1, n, \\ \text{and} & m \geq 0, \quad \xi_i \geq 0, \quad i = 1, n, \end{array} \right.$$

for $p = 1$ the dual formulation is:

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top G \alpha \\ \text{with} & \alpha^\top \mathbf{y} = 0 \text{ et } 0 \leq \alpha_i \leq \frac{1}{n} \quad i = 1, n \\ \text{and} & \nu \leq \alpha^\top \mathbb{I} \end{array} \right.$$

$$C = \frac{1}{m}$$

SVM with non symmetric costs

problem in the primal

$$\left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \|f\|_{\mathcal{H}}^2 + C^+ \sum_{\{i|y_i=1\}} \xi_i^p + C^- \sum_{\{i|y_i=-1\}} \xi_i^p \\ \text{with} & y_i(f(\mathbf{x}_i) + \alpha_0) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, n \end{array} \right.$$

for $p = 1$ the dual formulation is the following:

$$\left\{ \begin{array}{ll} \max_{\alpha \in \mathbb{R}^n} & -\frac{1}{2} \alpha^\top G \alpha + \alpha^\top \mathbb{I} \\ \text{with} & \alpha^\top \mathbf{y} = 0 \text{ and } 0 \leq \alpha_i \leq C^+ \text{ or } C^- \quad i = 1, n \end{array} \right.$$

Generalized SVM

$$\min_{f \in \mathcal{H}, \alpha_0 \in \mathbb{R}} \sum_{i=1}^n \max(0, 1 - y_i(f(\mathbf{x}_i) + \alpha_0)) + \frac{1}{C} \varphi(f)$$

φ convex

in particular $\varphi(f) = \|f\|_p^p$ with $p = 1$ leads to L1 SVM.

$$\left\{ \begin{array}{ll} \min_{\alpha \in \mathbb{R}^n, \alpha_0, \xi} & \mathbb{1}^\top \boldsymbol{\beta} + C \mathbb{1}^\top \boldsymbol{\xi} \\ \text{with} & y_i \left(\sum_{j=1}^n \alpha_j k(\mathbf{x}_i, \mathbf{x}_j) + \alpha_0 \right) \geq 1 - \xi_i, \\ \text{and} & -\beta_i \leq \alpha_i \leq \beta_i, \quad \xi_i \geq 0, \quad i = 1, n \end{array} \right.$$

with $\boldsymbol{\beta} = |\boldsymbol{\alpha}|$. the dual is:

$$\left\{ \begin{array}{ll} \max_{\gamma, \delta, \delta^* \in \mathbb{R}^{3n}} & \mathbb{1}^\top \boldsymbol{\gamma} \\ \text{with} & \mathbf{y}^\top \boldsymbol{\gamma} = 0, \quad \delta_i + \delta_i^* = 1 \\ & \sum_{j=1}^n \gamma_j k(\mathbf{x}_i, \mathbf{x}_j) = \delta_i - \delta_i^*, \quad i = 1, n \\ \text{and} & 0 \leq \delta_i, 0 \leq \delta_i^*, \quad 0 \leq \gamma_i \leq C, \quad i = 1, n \end{array} \right.$$

SVM reduction (reduced set method))

- objective: compile the model

- $f(x) = \sum_{i=1}^{n_s} \alpha_i k(\mathbf{x}_i, \mathbf{x}), n_s \ll n, \quad n_s \text{ too big}$

- compiled model as the solution of: $g(\mathbf{x}) = \sum_{i=1}^{n_c} \beta_i k(\mathbf{z}_i, \mathbf{x}), n_c \ll n_s$

- $\beta, \mathbf{z}_i$ and c are tuned by minimizing:

$$\min_{\beta, \mathbf{z}_i} \|g - f\|_H^2$$

where

$$\min_{\beta, \mathbf{z}_i} \|g - f\|_H^2 = \alpha^\top K_x \alpha + \beta^\top K_z \beta - 2\alpha^\top K_{xz} \beta$$

some authors advice $0,03 \leq \frac{n_c}{n_s} \leq 0,1$

- solve it by using use (stochastic) gradient (its a RBF problem)

SVM and probabilities (1/2)

$$\log \frac{\mathbb{P}(Y = 1|\mathbf{x})}{\mathbb{P}(Y = -1|\mathbf{x})} \text{ as (almost) the same sign as } f(\mathbf{x})$$

$$\log \frac{\mathbb{P}(Y = 1|\mathbf{x})}{\mathbb{P}(Y = -1|\mathbf{x})} = a_1 f(\mathbf{x}) + a_2 \quad \mathbb{P}(Y = 1|\mathbf{x}) = 1 - \frac{1}{1 + \exp^{a_1 f(\mathbf{x}) + a_2}}$$

a_1 et a_2 estimated using maximum likelihood

some facts

- SVM is universally consistent (coverges towards the Bayes risk)
- SVM asymptotically implements the bayes rule
- but theoretically: **no consistency towards conditional probabilities** (due to the nature of sparsity)
- to estimate conditional probabilities on an interval (typically $[\frac{1}{2} - \eta, \frac{1}{2} + \eta]$) to sparseness in this interval (all data points have to be support vectors)

SVM and probabilities (2/2)

An alternative approach

$$g(\mathbf{x}) - \varepsilon^-(\mathbf{x}) \leq \mathbb{P}(Y = 1|\mathbf{x}) \leq g(\mathbf{x}) + \varepsilon^+(\mathbf{x})$$

with $g(\mathbf{x}) = \frac{1}{1+4^{-f(\mathbf{x})-\alpha_0}}$

non parametric functions ε^- and ε^+ have to verify:

$$\begin{aligned} g(\mathbf{x}) + \varepsilon^+(\mathbf{x}) &= \exp^{-a_1(1-f(\mathbf{x})-\alpha_0)_+ + a_2} \\ 1 - g(\mathbf{x}) - \varepsilon^-(\mathbf{x}) &= \exp^{-a_1(1+f(\mathbf{x})+\alpha_0)_+ + a_2} \end{aligned}$$

with $a_1 = \log 2$ and $a_2 = 0$

logistic regression and the import vector machine

- Logistic regression is NON sparse
- kernalize it using the *dictionary* strategy
- Algorithm:
 - ▶ find the solution of the KLR using only a subset $\mathcal{S}$ of the data
 - ▶ build $\mathcal{S}$ iteratively using active constraint approach
- this trick brings sparsity
- it estimates probability
- it can naturally be generalized to the multiclass case

- efficient when uses:
 - ▶ a few import vectors
 - ▶ component-wise update procedure

- extention using L_1 KLR

Multiclass SVM

- one vs all: winner takes all
- one vs one:
 - ▶ max-wins voting
 - ▶ pairwise coupling: use probability
- global approach (size $c \times n$),
 - ▶ formal (different variations)

$$\left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \sum_{\ell=1}^c \|f_{\ell}\|_{\mathcal{H}}^2 + \frac{C}{p} \sum_{i=1}^n \sum_{\ell=1, \ell \neq y_i}^c \xi_{i\ell}^p \\ \text{with} & y_i (f_{y_i}(\mathbf{x}_i) + b_{y_i} - f_{\ell}(\mathbf{x}_i) - b_{\ell}) \geq 1 - \xi_{i\ell} \\ \text{and} & \xi_{i\ell} \geq 0 \text{ for } i = 1, \dots, n; \ell = 1, \dots, c; \ell \neq y_i \end{array} \right.$$

non consistent estimator but practically useful

- ▶ structured outputs

approach	problem size	number of sub problems
<i>all together</i>	$n \times c$	1
<i>1 vs. all</i>	n	c
<i>1 vs. 1</i>	$\frac{2n}{c}$	$\frac{c(c-1)}{2}$

Multiclass SVM

- one vs all: winner takes all
- one vs one:
 - ▶ max-wins voting
 - ▶ pairwise coupling: use probability – best results
- global approach (size $c \times n$),
 - ▶ formal (different variations)

$$\left\{ \begin{array}{ll} \min_{f \in \mathcal{H}, \alpha_0, \xi \in \mathbb{R}^n} & \frac{1}{2} \sum_{\ell=1}^c \|f_{\ell}\|_{\mathcal{H}}^2 + \frac{C}{p} \sum_{i=1}^n \sum_{\ell=1, \ell \neq y_i}^c \xi_{i\ell}^p \\ \text{with} & y_i (f_{y_i}(\mathbf{x}_i) + b_{y_i} - f_{\ell}(\mathbf{x}_i) - b_{\ell}) \geq 1 - \xi_{i\ell} \\ \text{and} & \xi_{i\ell} \geq 0 \text{ for } i = 1, \dots, n; \ell = 1, \dots, c; \ell \neq y_i \end{array} \right.$$

non consistent estimator but practically useful

- ▶ structured outputs

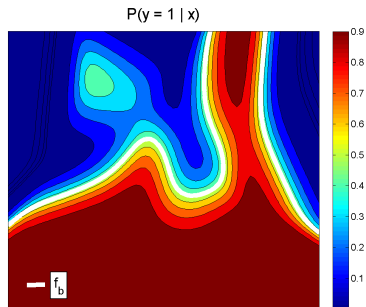
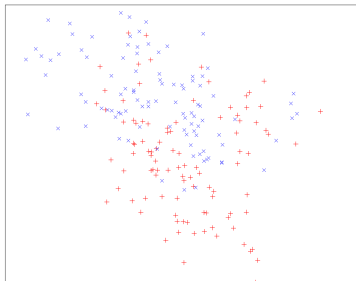
approach	problem size	number of sub problems
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Roadmap

1 Kernel machines

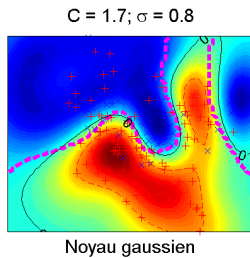
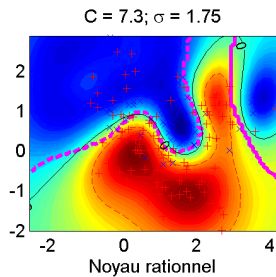
- Non sparse kernel machines
- sparse kernel machines: SVM
- Sparse kernel machines for regression: SVR
- practical SVM

Mixture data



- $x : 200 \times 2$
- $y : 100$ for each class
- mixture model with 10 gaussians
- the bayes error is known

the kernel effect

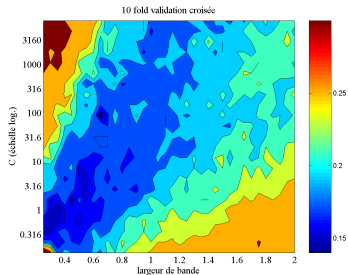
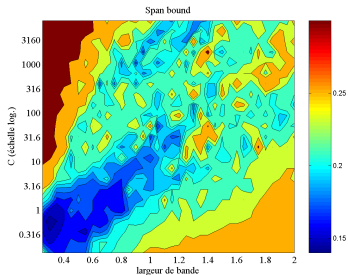
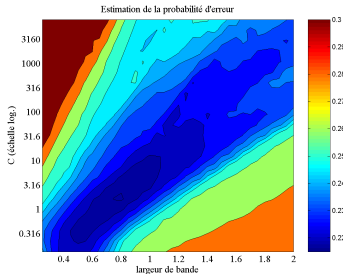


tuning C and σ : *grid search*

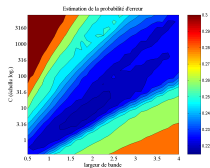
for $\sigma = 0.5 : 0.25 : 2$

for $C = 0.1$ à 10000

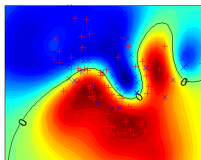
3 different error estimate



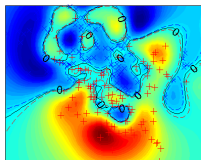
C and σ influence



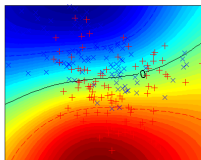
$C = 1; \sigma = 1$



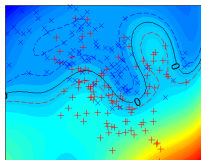
$C = 10000; \sigma = 1$



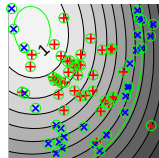
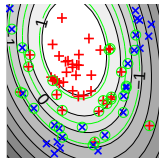
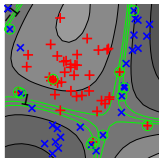
$C = 1; \sigma = 5$



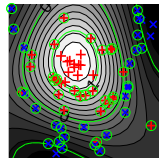
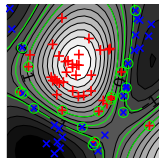
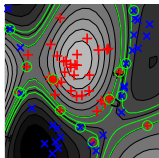
$C = 10000; \sigma = 5$



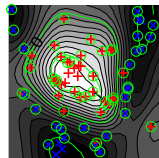
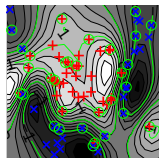
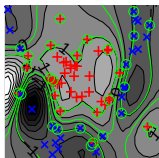
b grand



b moyen



b petit



C grand

C moyen

C petit

+

classe 1

x

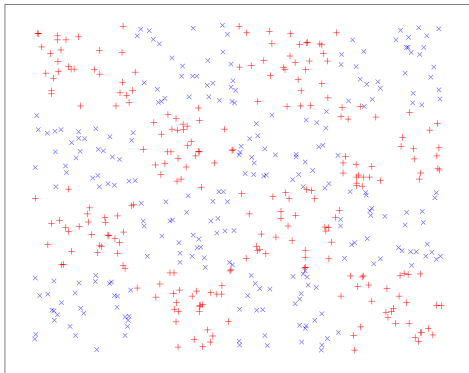
classe 2

o

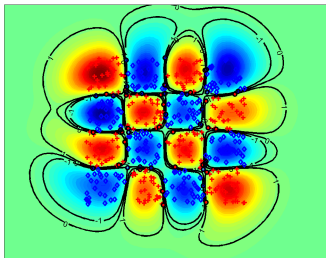
vecteur support

checker board

- 2 classes
- 500 examples
- separable

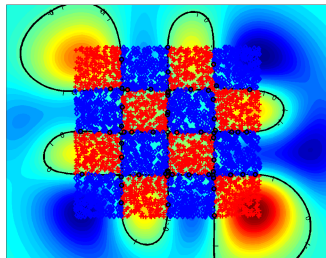


a separable case

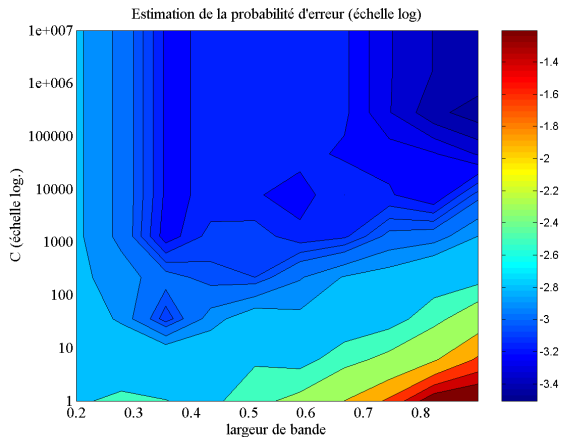


$n = 500$ data points

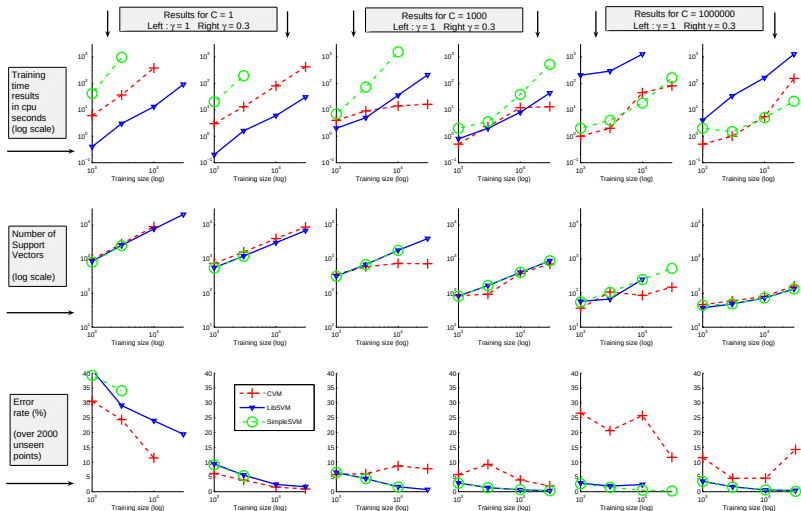
$n = 5000$ data points



tuning C and σ : *grid search*



empirical complexity



Historical perspective on kernel machines

statistics

1960 Parzen, Nadaraya Watson

1970 Splines

1980 Kernels: Silverman, Hardle...

1990 sparsity: Donoho (pursuit),
Tibshirani (Lasso)...

Statistical learning

1985 Neural networks:

- ▶ non linear - universal
- ▶ structural complexity
- ▶ non convex optimization

1992 Vapnik et. al.

- ▶ theory - regularization - consistency
- ▶ convexity - Linearity
- ▶ **Kernel** - universality
- ▶ **sparsity**
- ▶ results: MNIST

what's new since 1995

- Applications

- ▶ kernlisation $w^\top \mathbf{x} \rightarrow \langle f, k(\mathbf{x}, \cdot) \rangle_{\mathcal{H}}$
- ▶ kernel engineering
- ▶ sturtured outputs
- ▶ applications: image, text, signal, bio-info...

- Optimization

- ▶ dual: mloss.org
- ▶ regularization path
- ▶ approximation
- ▶ primal

- Statistic

- ▶ proofs and bounds
- ▶ model selection
 - ★ span bound
 - ★ multikernel: tuning (k and σ)

challenges: towards tough learning

- the size effect
 - ▶ ready to use: automatization
 - ▶ adaptative: on line context aware
 - ▶ beyond kenrels
- Automatic and adaptive model selection
 - ▶ variable selection
 - ▶ kernel tuning (k et σ)
 - ▶ hyperparametres: C , duality gap, λ
- $\mathbb{P}$ change
- Theory
 - ▶ non positive kernels
 - ▶ a more general representer theorem

biblio: kernel-machines.org

- John Shawe-Taylor and Nello Cristianini Kernel Methods for Pattern Analysis, Cambridge University Press, 2004
- Bernhard Schölkopf and Alex Smola. Learning with Kernels. MIT Press, Cambridge, MA, 2002.
- Trevor Hastie, Robert Tibshirani and Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, springer, 2001
- Léon Bottou, Olivier Chapelle, Dennis DeCoste and Jason Weston Large-Scale Kernel Machines (Neural Information Processing, MIT press 2007
- Olivier Chapelle, Bernhard Scholkopf and Alexander Zien, Semi-supervised Learning, MIT press 2006
- Vladimir Vapnik. Estimation of Dependences Based on Empirical Data. Springer Verlag, 2006, 2nd edition.
- Vladimir Vapnik. The Nature of Statistical Learning Theory. Springer, 1995.
- Grace Wahba. Spline Models for Observational Data. SIAM CBMS-NSF Regional Conference Series in Applied Mathematics vol. 59, Philadelphia, 1990
- Alain Berlinet and Christine Thomas-Agnan, Reproducing Kernel Hilbert Spaces in Probability and Statistics, Kluwer Academic Publishers, 2003
- Marc Attéia et Jean Gaches , Approximation Hilbertienne - Splines, Ondelettes, Fractales, PUG, 1999